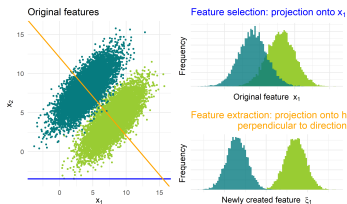
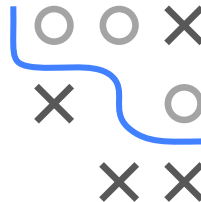


Feature Selection: Introduction



- Too many features can be harmful in prediction
- Selection vs. extraction
- Types of selection methods

INTRODUCTION I

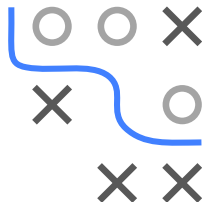
Feature selection:

Finding a well-performing, hopefully small set of features for a task.

Feature selection is critical for

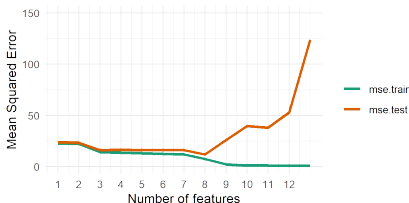
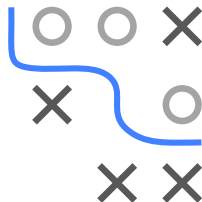
- reducing noise and overfitting
- improving performance/generalization
- enhancing interpretability by identifying most informative features

Features can be selected based on domain knowledge, or data-driven algorithmic approaches. We focus on the latter here.



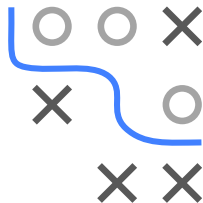
MOTIVATION I

- Naive view:
 - More features $\rightarrow$ more information $\rightarrow$ discriminant power $\uparrow$
 - Model is not harmed by irrelevant features since their parameters can simply be estimated as 0.
- In practice, irrelevant and redundant features can “confuse” learners (see **curse of dimensionality**) and worsen performance.
- Example: In linear regression, R^2 is monotonically increasing in p , but adding irrelevant features leads to overfitting (capturing noise).



MOTIVATION II

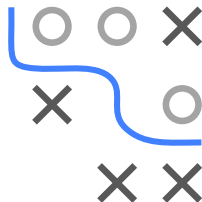
- In high-dimensional data sets, we often have prior information that many features are either irrelevant or of low quality
- Having redundant features can cost something during prediction (money or time)
- Many models require $n > p$ data. Thus, we either need to
 - adapt models to high-dimensional data (e.g., regularization)
 - design entirely new procedures for $p > n$ data
 - use filter preprocessing methods from this lecture



SIZE OF DATASETS I

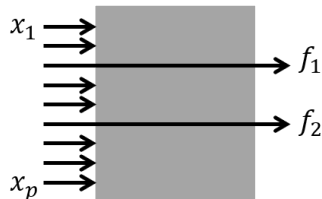
Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

- **Classical setting:** Up to around 10^2 features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality:** At around 10^2 to 10^3 features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data:** 10^3 to 10^9 or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called $p \gg n$.



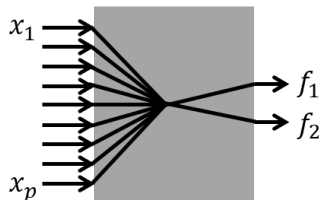
FEATURE SELECTION VS. EXTRACTION I

Feature selection

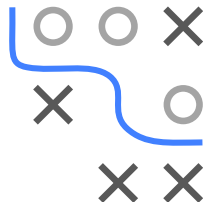


- Creates a subset of original features $\mathbf{x}$ by selecting $\tilde{p} < p$ features $\mathbf{f}$.
- Retains information on selected individual features.

Feature extraction

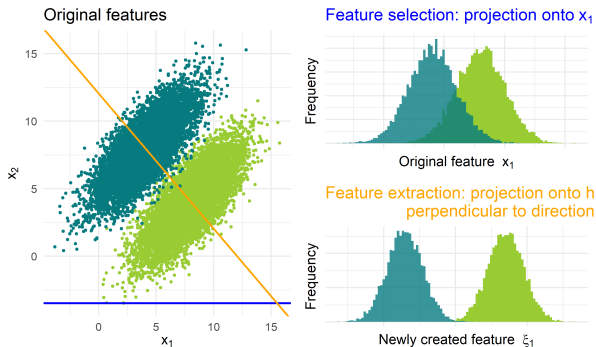
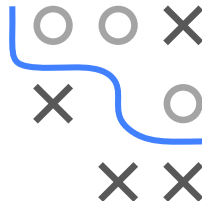


- Maps p features in $\mathbf{x}$ to $\tilde{p}$ extracted features $\mathbf{f}$.
- Info on individual features can be lost through (non-)linear combination.



FEATURE SELECTION VS. EXTRACTION II

- Both FS and FE contribute to
 - 1) dimensionality reduction and 2) simplicity of models
- FE can be unsupervised (PCA, multidim scaling, manifold learning) or supervised (supervised PCA, partial least squares)
- FE can produce lower dim projections which can work better than FS; whether FE+model is interpretable depends on how interpretable extracted features are



TYPES OF FEATURE SELECTION METHODS I

In rest of the chapter, we introduce different types of methods for FS:

Example: embedded method (Lasso) regularizing model params with L_1 penalty enables “automatic” feature selection:

